Nine Soft SF CLOSED/PROVED lineage items - mathematical statements

Outside original Residual Ternary Algebra and Bridge C papers

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Mode: DRAFT-ONLY from already-captured FINDINGS. NO computation. REPORT only - not a new paper.
No delegation. Automorphous gate STANDS. tip Soft SF LOCK 7cddb188... tip_touched NO. Kerr BLOCKED.
Parent: PHYSICIST_REPORT_PROOFS_OUTSIDE_ORIGINAL_RTA_BRIDGE_C_2026-10-27.md (sha 86975f8c...).

Method: each Mathematical statement is quoted or faithfully paraphrased from captured FINDINGS STATUS / Claim / Conclusion lines only. Rows 4 and 8 remain PARTIAL in those FINDINGS.

1. INV-Ind

Captured Soft SF status: PROVED Soft SF; invent-pending DISCHARGED

Mathematical statement.

Let G = GL_2(Q_p), B = P the upper-triangular Borel, and chi : P -> F_p-bar^x a smooth character inflated through the torus T. Then c-Ind_B^G chi = Ind_B^G chi, because P\G is compact (Iwasawa G = PK), so the compact-mod-P support condition is automatic; and Ind_B^G chi = {f : G -> F_p-bar smooth : f(bg) = chi(b) f(g)} with right-translation G-action - exactly the INV-Ind convention.

2. GAP-SG-K1

Captured Soft SF status: CLOSED Soft SF; NOT OPERATIVE

Mathematical statement.

Let K = GL_2(O_F), K_1 = ker(K -> Gamma) the first congruence subgroup (Gamma isomorphic to GL_2(F_p)), and pi any smooth representation over F_p. Then every irreducible smooth K-subrepresentation of pi lies in pi^{K_1}, and soc_K pi = soc_K(pi^{K_1}) = soc_Gamma(pi^{K_1}). In particular, for the principal series pi = pi_alpha, soc_K(pi_alpha) = soc_Gamma(pi_alpha^{K_1}).

3. Colmez18 D1+D2 / beta <-> rho

Captured Soft SF status: D1+D2 CLOSED Soft SF; beta <-> rho CLOSED Soft SF; NOT OPERATIVE

Mathematical statement.

(D1) With BM's pi_alpha := Ind_P^G omega tensor omega^{-1} and Colmez's B(delta_1, delta_2) = Ind_B delta_2 tensor delta_1 omega^{-1}, one has B(1, omega) = Ind_B omega tensor omega^{-1} = pi_alpha. (D2) Colmez's V(omega, 1, tau^perp) is the nonsplit extension 0 -> omega -> V -> 1 -> 0; under the BM <-> Colmez dictionary this matches BM's nonsplit 0 -> omega -> rho -> 1 -> 0, and V(beta) congruent to rho.

4. filtr invent-prove (parent) / CANDIDATE (10)

Captured Soft SF status: Soft SF PARTIAL (w <= 10 and w != 2,4 PROVED Soft SF; w notin {6,8} was OPEN)

Mathematical statement.

Let w = w(theta^3 f) be the filtration weight of theta^3 f (mod-7 ambient of parent FINDINGS). Soft SF proved w(theta^3 f) <= 10, w(theta^3 f) != 2, and w(theta^3 f) != 4 (odd weights trivial), leaving open whether w is in {6,8}.

5. GAP-FILTR-68

Captured Soft SF status: CLOSED Soft SF; w(theta^3 f) notin {6,8} PROVED Soft SF

Mathematical statement.

Continuing the parent PARTIAL: w(theta^3 f) notin {6,8}. Combined with w <= 10, w != 2,4, and odd weights trivial, under the filtr-invent-pick L14 convention one obtains w(theta^3 f) = 10.

6. GAP-SK-p3-1

Captured Soft SF status: CLOSED Soft SF (published source on disk)

Mathematical statement.

Let rho be absolutely irreducible with rho|I congruent to (omega_2^{r+1} oplus omega_2^{p(r+1)}) tensor omega^m, so pi congruent to pi(r,0,chi) = (ind_{KZ}^G Sym^r / T) tensor chi o det (Breuil I). Then soc_K pi congruent to (Sym^r oplus Sym^{p-1-r} tensor det^r) tensor det^m. (Valid for all p >= 3; includes p = 3.)

7. GAP-SK-p3-1a

Captured Soft SF status: CLOSED Soft SF; citation FLAG discharged; NOT OPERATIVE

Mathematical statement.

(Step 2 of GAP-SK-p3-1.) Every irreducible smooth F_p-bar[K]-representation, K = GL_2(Z_p), is of the form sigma congruent to Sym^s F_p-bar^2 tensor det^a inflated from GL_2(F_p), for some 0 <= s <= p-1 and a mod (p-1), and dim sigma^{I(1)} = 1.

8. GAP-SK-p3-1b

Captured Soft SF status: PARTIAL Soft SF; NOT OPERATIVE; Brauer theorem UNQUOTED on disk

Mathematical statement.

The number of p-regular conjugacy classes in GL_2(F_p) equals p(p-1), matching the label count of {Sym^s tensor det^a : 0 <= s <= p-1, a mod (p-1)}. Soft SF proved this count equality; completeness of the irreducible list would follow if Brauer's theorem (#irreducibles = #p-regular classes) were granted, but that theorem remained unquoted on disk - hence PARTIAL.

9. GAP-SK-p3-1c (+ 1c-MC)

Captured Soft SF status: 1c CLOSED Soft SF Brauer-free; 1c-MC ALL_OK (p=3,5,7); NOT OPERATIVE

Mathematical statement.

(1c) Every irreducible F_p-bar-representation V of G = GL_2(F_p) is isomorphic to Sym^s tensor det^a for some 0 <= s <= p-1 and a mod (p-1) (Brauer-free route via nonzero U-coinvariants and induction from the Borel). (1c-MC) For p in {3,5,7}, machine check ALL_OK: every (s,a) occurs as a socle constituent of some Ind_B^G psi, and socle dimensions / U-weights match the Sym-list (discharges remaining FLAG).

Explicit non-claims

This PDF is a report of captured mathematical statements, not a new paper and not a tip fold. No computation was run for this deliverable beyond drafting from already-captured FINDINGS and producing this PDF. Rows 4 and 8 remain PARTIAL in captured STATUS; included because they are among the nine lineage rows named in the parent draft report section A. No delegation. Everyone else stays stopped until automorphous.